

M7_Assignment

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Q2 Electricity Usage

Q2.1 Loading Packages and Data

Load the fpp3 and tidyverse packages. Save the data into an object called Usage.

```
library(fpp3)
```

```
Warning: package 'fpp3' was built under R version 4.4.2
```

```
Registered S3 method overwritten by 'tsibble':
```

```
  method          from  
as_tibble.grouped_df dplyr
```

```
-- Attaching packages ----- fpp3 1.0.1 --
```

```
v tibble      3.2.1    v tsibble      1.1.5  
v dplyr       1.1.4    v tsibbledata 0.4.1  
v tidyr       1.3.1    v feasts      0.4.1  
v lubridate   1.9.3    v fable       0.4.1  
v ggplot2     3.5.1
```

```
Warning: package 'tsibble' was built under R version 4.4.2
```

```
Warning: package 'tsibbledata' was built under R version 4.4.2
```

```
Warning: package 'feasts' was built under R version 4.4.2
```

Warning: package 'fabletools' was built under R version 4.4.2

Warning: package 'fable' was built under R version 4.4.2

```
-- Conflicts ----- fpp3_conflicts --
x lubridate::date()      masks base::date()
x dplyr::filter()        masks stats::filter()
x tsibble::intersect()   masks base::intersect()
x tsibble::interval()    masks lubridate::interval()
x dplyr::lag()            masks stats::lag()
x tsibble::setdiff()      masks base::setdiff()
x tsibble::union()        masks base::union()
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v forcats 1.0.0      v readr 2.1.5
v purrr 1.0.2        v stringr 1.5.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter()      masks stats::filter()
x tsibble::interval()  masks lubridate::interval()
x dplyr::lag()          masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
Usage <-read_csv("https://jagelves.github.io/Data/ElectricityBill.csv")
```

Rows: 38 Columns: 6

```
-- Column specification -----
Delimiter: ","
```

dbl (6): Year, Month, Meter Read, Usage (kWh), Avg. Daily Usage, Bill Amount

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Q2.2 Date Variables

The usage tibble has no date variable. Create a variable called date using the mutate(), yearmonth(), and seq() functions. To do this, wrap the seq() function with the yearmonth() function (i.e., the seq() function is inside the yearmonth() function). The arguments for the seq() function are as follows:

- set the from argument to ymd("2021-06-01")
- set the length.out argument to 38
- set the by argument to "1 month"

```
Usage %>%  
  mutate(date = yearmonth(seq(from = ymd("2021-06-01"), length.out = 38, by = "1 month")))
```

Q2.3 Tsibble

Use the usage tibble to create a tsibble that uses date as the index. Use piping to:

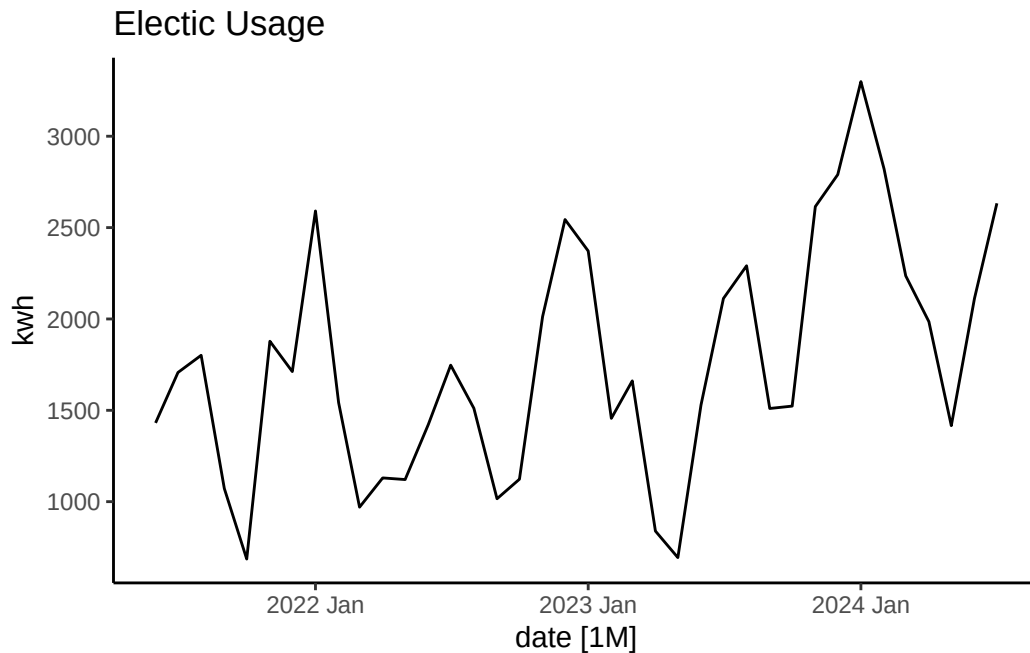
Rename the Usage (kWh) to kwh Select the date and kwh variables. Store the result in an object called usage_ts using the assignment operator ->

```
usage %>%  
  rename(kwh = `Usage (kWh)`) %>%  
  select(date, kwh) %>%  
  as_tsibble(index=date) -> usage_ts
```

Q2.4 Plot

Create a plot of the kWh variable in the usage_ts tsibble. Include a title that reads “Electric Usage” and use the theme_classic() function. Make no other modifications and upload a screenshot below:

```
usage_ts %>%  
  autoplot(kwh) + theme_classic() +  
  labs(title = "Electric Usage")
```



Q2.5 Test and Train

Create test and training sets using the `filter_index()` function. The training set should start in July 2021 and end in April 2023. The rest of the data should be in the test set. Include the code below:

```
usage_ts %>% filter_index("July 2021"~"April 2023") -> train
usage_ts %>% filter_index("May 2023"~.) -> test
```

Q2.6 Fit

Create an object called `fit` that estimates the `ARIMA()`, `ETS()`, and a `TSLM()` model with `trend()` and `seasonality()` using the training set. No arguments (other than `kwh`) should be passed in the `ARIMA` or `ETS` models to allow the algorithm to choose the best model. Include your code below:

```
fit <- model(train, ARIMA=ARIMA(kwh),
             ETS=ETS(kwh),
             LS=TSLM(kwh~trend()+season()))
```

Q2.7 Accuracy

Use the fit object to retrieve accuracy measures of the forecast relative to the test set (no cross-validation). Arrange the table in ascending order by RMSE. Upload a screenshot of the table below:

```
fit %>% forecast() %>% accuracy(test) %>% arrange(RMSE)
```

Warning: The future dataset is incomplete, incomplete out-of-sample data will be treated as missing. 9 observations are missing between 2024 Aug and 2025 Apr

```
# A tibble: 3 x 10
  .model .type    ME  RMSE   MAE   MPE  MAPE  MASE  RMSSE  ACF1
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 LS     Test   418.  567.  492.  14.5  25.1   NaN   NaN  0.402
2 ARIMA  Test   595.  859.  685.  18.9  30.6   NaN   NaN  0.477
3 ETS    Test   563.  866.  701.  15.4  33.4   NaN   NaN  0.451
```

Q2.8 Information Criterion

Use the fit object and the glance() function to retrieve the AIC, AICc and the BIC. Upload a screenshot of the output below:

```
fit %>%
  glance() %>%
  select(".model", "AIC", "AICc", "BIC")
```

```
# A tibble: 3 x 4
  .model  AIC  AICc  BIC
  <chr>  <dbl> <dbl> <dbl>
1 ARIMA  340.  342.  344.
2 ETS    349.  350.  352.
3 LS     257.  317.  272.
```

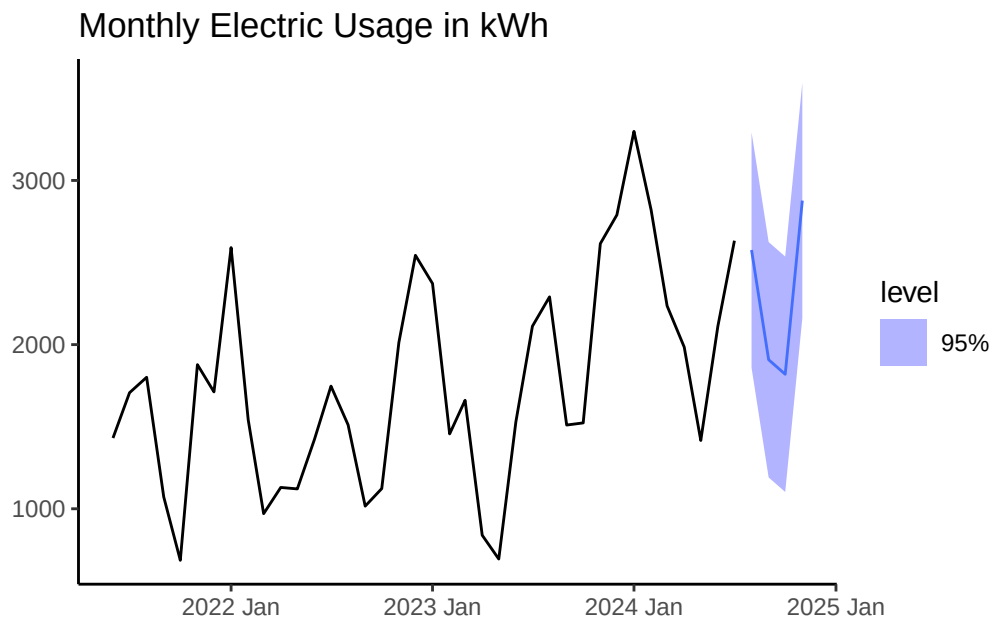
Q2.9

Use piping to complete the following steps. Start with the usage_ts object.

Use the model() function to estimate the best model according to question Q2.7 and Q2.8. forecast four periods ahead (h=4) autoplot() with level=95 autolayer() with usage_ts, title="Monthly Electric Usage in kWh", theme_classic(), and no captions for y or x axis.

```
usage_ts %>%
  model(LS=TSLM(kwh~trend()+season())) %>%
  forecast(h=4) %>%
  autoplot(level=95) +
  theme_classic() +
  autolayer(usage_ts) +
  labs(title = "Monthly Electric Usage in kWh",
        y="", x="")
```

Plot variable not specified, automatically selected ``vars = kwh``



Q3 Update Forecast

Update the Netflix data up to the current date by browsing the web. Use a time series forecasting method learned in class to provide a forecast of Netflix world-wide subscribers for the upcoming quarter.

```
#Update Netflix data from web.
Netflix <- read_csv("https://jagelvels.github.io/Data/Netflix.csv")
```

```

Rows: 46 Columns: 2
-- Column specification -----
Delimiter: ","
chr (1): Quarter
dbl (1): Subscribers

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```

Netflix %>% add_row(Quarter="2024:Q3",Subscribers=282.72) -> Netflix
tail(Netflix)

```

```

# A tibble: 6 x 2
  Quarter Subscribers
  <chr>      <dbl>
1 2023:Q2      238.
2 2023:Q3      247.
3 2023:Q4      260.
4 2024:Q1      270.
5 2024:Q2      278.
6 2024:Q3      283.

```

```

Netflix %>%
  mutate(Quarter=yearquarter(Quarter)) %>%
  as_tsibble(index=Quarter) -> Netflix_ts

```

```

Netflix_ts %>% features(Subscribers, features = c(unitroot_kpss, unitroot_ndiffs))

```

```

# A tibble: 1 x 3
  kpss_stat kpss_pvalue ndiffs
  <dbl>      <dbl>    <int>
1     1.27      0.01      1

```

```

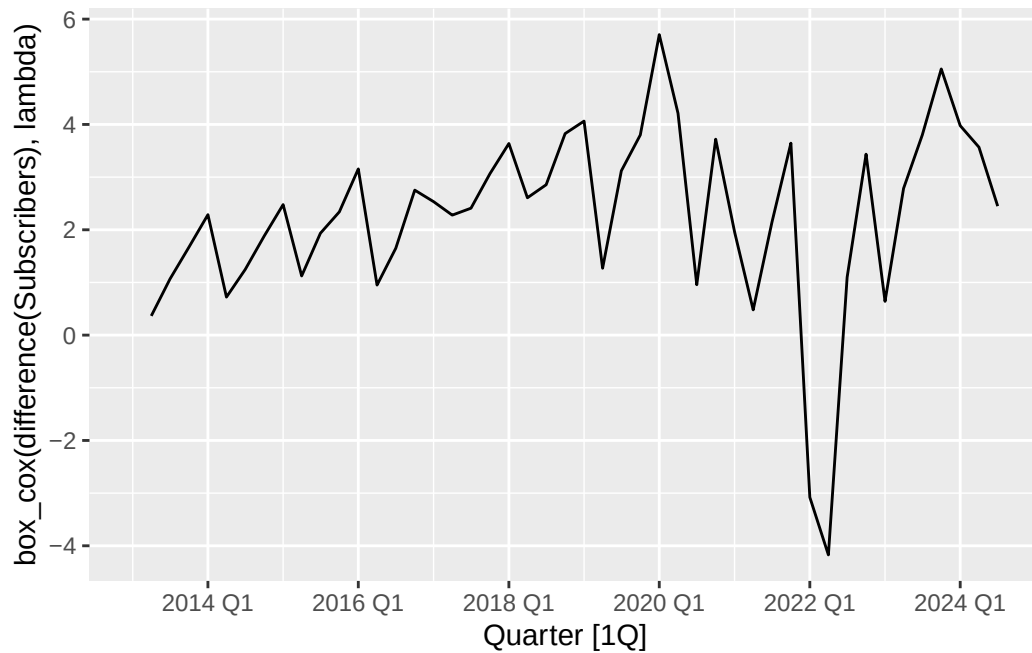
Netflix_ts %>%
  features(Subscribers, features = guerrero) %>% pull(lambda_guerrero) -> lambda
Netflix_ts %>% autoplot(box_cox(difference(Subscribers), lambda))

```

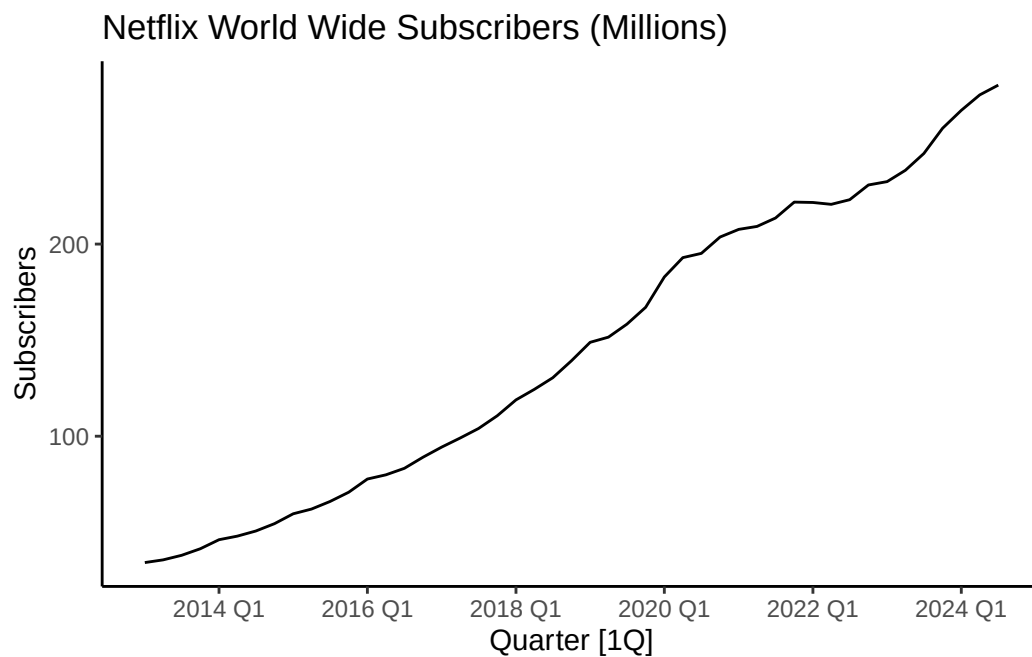
```

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_line()`).

```



```
Netflix_ts %>%
  autoplot(Subscribers) + theme_classic() +
  labs(title = "Netflix World Wide Subscribers (Millions)")
```



```
train_net <- filter_index(.data=Netflix_ts, ~"2021 Q4")
test_net <- filter_index(.data=Netflix_ts, "2022 Q1"~.)
```

```
fit <- model(Netflix_ts, DAMPEDS=ETS(Subscribers ~ error("A") + trend("Ad") + season("A")))
fit %>% forecast(h=1)
```

```
# A fable: 1 x 4 [1Q]
# Key:      .model [1]
#   .model Quarter
#   <chr>      <qtr>
# 1 DAMPEDS 2024 Q4
# i 2 more variables: Subscribers <dist>, .mean <dbl>
```

```
fit %>% glance()
```

```
# A tibble: 1 x 9
#   .model sigma2 log_lik  AIC  AICc  BIC  MSE  AMSE  MAE
#   <chr>   <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
# 1 DAMPEDS  9.34    -138.  296.  302.  314.  7.55  29.1  2.04
```